

GATE-LEVEL SIMULATIONS: An Increasing Trend

Gate-level simulation (GLS) is used to boost the confidence regarding implementation of a design and can help verify dynamic circuit behaviour, which cannot be verified accurately by static methods. It is a significant step in the verification process.

GLS overcomes the limitations of static-timing analysis and is increasing being used due to low power issues, complex timing checks at 40nm and below, design for test (DFT) insertion at gate level and low power considerations. For DFT, scan chains are inserted after the gate-level netlist is created; GLS is often used to determine whether scan chains are correct.

Technology libraries at 45nm and below have far more timing checks and complex timing checks than older process nodes. GLS may take up to one-third of the simulation time and could potentially take most of the debugging time. It is run after RTL code is simulated and synthesised into a gate-level netlist. It requires a complete reset of the design.

Reasons for running GLS are reset verification, X optimism in RTL, timing verification on multi-cycle/asynchronous

paths and basic heartbeat test.

In reset verification, GLS can verify system initialisation and show that the reset sequence is correct.

In X optimism in RTL, an RTL simulator may optimistically assign zero or one to a value that a GLS would identify as X (unknown).

In timing verification on multi-cycle/asynchronous paths, static-timing analysis cannot identify asynchronous interfaces, and has constraint requirements that impact false and multi-cycle paths.

In the basic heartbeat test, some verification teams may want to run a limited sanity check to verify the functionality at the gate level. As GLS runs much more slowly than an RTL simulation, it potentially has significant impact on the verification closure cycle.

Cadence incise enterprise simulator has several features such as zero-delay simulation, built-in delay mode control functions to reduce simulation time, selectively disabling delays in sections of the model where timing is not currently a concern, detecting potential zero-delay gate loops, correcting race conditions that occur in zero-delay mode, disabling timing checks for the entire simulation or for selected blocks, controlling the number of timing check violations, using multi-snapshot incremental elaboration to improve elaboration performance, using wave dumping only if required, avoid or use selectively command-line options that provide additional information and access to objects for debugging.

Incise also offers a timing file that lets you turn off the timing for particular instances in a design. Palladium XP accelerator/emulator can offer speeds 10,000 times faster than simulation. If full debug access is needed, a switch can provide it. There is also an option (-ZLIB) that can compress snapshots and save disk space, while letting users set the level of compression.



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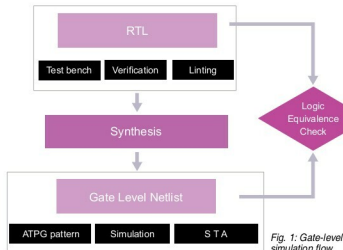


Fig. 1: Gate-level simulation flow